Example Program for using Custom Exception

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Example 1:

Create a custom exception which will check the age of voter and if below 18 will

```
class InvalidAgeException extends Exception { private final int age; public
InvalidAgeException(String message, int age) { super(message); this.age = age;
} public int getAge() { return age; } } class Voter { static void register(int ageN) throws
InvalidAgeException { if (ageN < 18) throw new InvalidAgeException("Must be 18 or older", ageN);
System.out.println("Registered successfully"); } } public class Main { public static void
main(String[] args) { try { Voter.register(25); // succeeds Voter.register(16); // throws } catch
(InvalidAgeException e) { System.out.println("Error: " + e.getMessage()); //displays the exception
message “Must be 18 or older” System.out.println("Age entered: " + e.getAge()); } } }
```

}

interface Payment {

void pay(double amount);

}

class CardPayment implements Payment {

public void pay(double amount) {

```
System.out.println("Paid " + amount + " by card");
```

}

}

class UpiPayment implements Payment {

public void pay(double amount) {

```
System.out.println("Paid " + amount + " by UPI");
```

}

}

```
public class InterfaceDemo {
public static void main(String[] args) {
Payment p = new CardPayment(); // reference of interface type
```

p.pay(500);

p = new UpiPayment(); // swap implementation, same code

p.pay(300);

}

}

```
interface X { default void hi() { System.out.println("X"); } }
interface Y { default void hi() { System.out.println("Y"); } }
```

class Z implements X, Y {

public void hi() { X.super.hi(); } // resolves the conflict

}

Using interface

Animal (interface)

/ \

Dog Cat (both override default sound())

\ /

Hybrid (must resolve the conflict)

interface Animal {

default void sound() {

```
System.out.println("Animal makes a sound");
```

}

}

interface Dog extends Animal {

@Override

default void sound() {

```
System.out.println("Dog barks");
```

}

}

interface Cat extends Animal {

@Override

default void sound() {

```
System.out.println("Cat meows");
```

}

}

```
// Diamond: Hybrid inherits sound() through Dog and through Cat
```

class Hybrid implements Dog, Cat {

@Override

public void sound() {

Dog.super.sound(); // explicitly choose Dog's version

}

}

```
public class DiamondDemo {
```

```
public static void main(String[] args) {
```

Hybrid h = new Hybrid();

h.sound();

```
}
}
```